

G3 Eleven-Week Roadmap

AMS Intake Platform - Algorithmic Intelligence Engine

Prepared by: Hugo Davion & Axel Brazeau
Group 3 — Algorithmic Intelligence Engine
<https://github.com/hodncgi/AMSSchatBot>

1. Introduction

This document describes the week-by-week delivery plan for Group 3, from project launch through final delivery. Each week includes goals, tasks, deliverables, and linked user stories. The plan covers backend algorithms, AI chat, and the onboarding chatbot UI without merging into an external site.

2. Eleven-Week Plan

Week 1 - Project launch and user needs

Goals:

- Define Group 3 scope with referent and Group 2.
- Identify user personas: new employee, intake consultant, manager.
- Draft user stories and high-level requirements.

Tasks:

- Kickoff meeting and role allocation (Hugo and Axel).
- Map onboarding page as sole chatbot entry point.
- List questions users must be able to ask (onboarding and risk analysis).
- Write first version of user stories (US-01 to US-16).

Deliverables:

- User stories v0.1
- Persona list
- Scope document

Linked user stories: US-01, US-11

Week 2 - Architecture and API design

Goals:

- Design backend architecture for NBQ, Change Risk, and Chat.
- Choose local AI stack (LM Studio + Mistral).
- Define how the chatbot uses all G3 backend routes.

Tasks:

- Architecture diagram: Onboarding UI to FastAPI to NBQ, Change Risk, Chat, Health.
- API contract draft for Group 2.
- Document chatbot knowledge scope (RAG and API-aware answers).
- LM Studio installation and model selection.

Deliverables:

- Architecture note
- API contract draft
- LM Studio ready

Linked user stories: US-08, US-12

Week 3 - Backend foundation

Goals:

- Create FastAPI project structure.
- Implement health monitoring.

Tasks:

- backend/ setup: main.py, config.py, Pydantic models.
- GET /health (API, LM Studio, RAG status).
- CORS for onboarding frontend (localhost:3000).
- Repository structure and README.

Deliverables:

- FastAPI on port 8000
- GET /health
- GitHub repo

Linked user stories: US-08

Week 4 - Next-Best Question (NBQ)

Goals:

- Implement NBQ algorithm and API.
- Add Mistral justification for why field.

Tasks:

- Weighting rules and QUESTION_BASE.
- POST /nbq/next implementation.
- generate_nbq_explanation() via LM Studio.
- Healthcare + security test scenario.

Deliverables:

- POST /nbq/next
- NBQ tests

- OpenAPI entry

Linked user stories: US-04, US-05

Week 5 - Change Risk analysis

Goals:

- Implement change risk scoring engine.
- Document decision-support scope.

Tasks:

- `compute_change_risk()` with drivers.
- POST `/change-risk` validation.
- Edge cases (score minimum 0).
- Clarify Group 3 vs AMS lead responsibilities in docs.

Deliverables:

- POST `/change-risk`
- API.md section
- Test cases

Linked user stories: US-03, US-06, US-11

Week 6 - Knowledge base and test bench

Goals:

- Build CGI knowledge for RAG.
- Provide G3 test interface for consultants.

Tasks:

- `knowledge_base.py` topics (onboarding, AMS, change risk, NBQ).
- `catalog.json` document chunks.
- Include change risk and NBQ explanations in knowledge.
- `frontend/g3-test` HTML bench.

Deliverables:

- ~40 chunks indexed
- GET `/g3-test`
- Risk and NBQ content in KB

Linked user stories: US-10, US-03, US-04, US-15

Week 7 - RAG and AI service (Mistral)

Goals:

- Implement retrieval and LM Studio integration.
- Prepare chat to answer from CGI knowledge.

Tasks:

- rag_service.py keyword retrieval.
- ai_service.py prompts and call_lm_studio.
- System prompt includes platform and risk analysis context.
- Model auto-detection and timeout configuration.

Deliverables:

- RAG pipeline
- LM Studio chat working
- ai_service.py

Linked user stories: US-01, US-07

Week 8 - Chat API and backend route awareness

Goals:

- Expose POST /chat for onboarding conversations.
- Ensure chatbot answers leverage full G3 backend (risk, NBQ, health).

Tasks:

- POST /chat with history and RAG injection.
- Enrich knowledge base with API behaviour (change-risk, nbq).
- Fallback mode when LM Studio offline.
- Pipeline metadata in responses (lm_studio, mode).

Deliverables:

- POST /chat
- Chat knows risk and NBQ APIs
- Fallback mode

Linked user stories: US-01, US-03, US-04, US-09, US-12, US-13, US-14

Week 9 - Onboarding chatbot UI

Goals:

- Build React chatbot on the Onboarding page only.

- Connect UI to /api/chat and display sources and AI status.

Tasks:

- ChatInterface on onboarding tab (not other tabs).
- Vite proxy to FastAPI.
- AiPipelineBar (LM Studio visibility).
- Suggested questions including risk and AMS topics.

Deliverables:

- Chatbot on port 3000
- Onboarding tab only
- Live Mistral answers

Linked user stories: US-01, US-02, US-07, US-12, US-13

Week 10 - Integration testing and documentation

Goals:

- End-to-end test campaign.
- Finalize setup and API documentation.
- Prepare Group 2 handover.

Tasks:

- Test onboarding questions, change risk explanations, NBQ questions.
- SETUP.md, API.md, AI.md updates.
- G2 API contract review.
- Manager progress reports.

Deliverables:

- Test report
- Full documentation
- Stable API v1.0

Linked user stories: US-08, US-09, US-10, US-14

Week 11 - Finalization and delivery

Goals:

- Polish onboarding chatbot and backend.
- Deliver demo package to management.
- Close Group 3 scope and plan v2.

Tasks:

- Bug fixes and UX polish on onboarding chatbot.
- Full demo script: onboarding chat, NBQ, change risk via test bench.
- start.ps1, .env.example, GitHub sync.
- v2 roadmap: continuity score, Group 1 relational database (US-15, US-16), semantic RAG.

Deliverables:

- Production-ready POC on AMSChatBot repo
- Manager presentation
- Final user stories validated (US-01 to US-16)

Linked user stories: US-01, US-05, US-06, US-12, US-15, US-16

3. Roadmap Overview

Week	Theme	Key deliverable	User stories
1	Project launch and user needs	User stories v0.1	US-01, US-11
2	Architecture and API design	Architecture note	US-08, US-12
3	Backend foundation	FastAPI on port 8000	US-08
4	Next-Best Question (NBQ)	POST /nbq/next	US-04, US-05
5	Change Risk analysis	POST /change-risk	US-03, US-06, US-11
6	Knowledge base and test bench	~40 chunks indexed	US-10, US-03, US-04, US-15
7	RAG and AI service (Mistral)	RAG pipeline	US-01, US-07
8	Chat API and backend route awareness	POST /chat	US-01, US-03, US-04, US-09, US-12, US-13, US-14
9	Onboarding chatbot UI	Chatbot on port 3000	US-01, US-02, US-07, US-12, US-13
10	Integration testing and documentation	Test report	US-08, US-09, US-10, US-14
11	Finalization and delivery	Production-ready POC on AMSChatBot repo	US-01, US-05, US-06, US-12, US-15, US-16

4. Backend Routes

Route	Purpose	Used by
POST /chat	Onboarding conversation (Mistral + RAG)	Onboarding chatbot

POST /change-risk	Change risk score 0-100 + drivers	Test bench, chatbot knowledge
POST /nbq/next	Next best intake question + why	Test bench, chatbot knowledge
GET /health	API, LM Studio, RAG status	Chatbot UI, operators

5. Conclusion

This roadmap delivers the Group 3 engine from user needs through algorithms, AI chat, and onboarding UI. Week 11 closes the scope with a management-ready demo and validated user stories. Version 2 adds Group 1 relational database integration (US-15, US-16), and semantic RAG when corporate network constraints allow.

Hugo Davion & Axel Brazeau
Group 3 — Algorithmic Intelligence Engine